

- (a) (i) With reference to Fig. 3.1, suggest **and** explain how the presence of glucose in the extracellular environment prevents lactose entering the bacterial cell.

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- (ii) Suggest the advantages for the bacterial cell in preventing the entry of lactose when glucose is present in the extracellular environment.

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- (iii) The glucose 6-phosphate in the cytoplasm of the bacterial cell is used to produce fructose 1,6-bisphosphate.

Fructose 1,6-bisphosphate is used in glycolysis.

Outline the events that occur in glycolysis following the production of fructose 1,6-bisphosphate.

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